

The Zero-Variance Fraction: Diagnosing and Budgeting Signal Starvation in Group-Relative Reinforcement Learning Post-Training of Large Language Models

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Master of Technology

by

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Declaration

I declare that this thesis presents my own work, carried out in the fourth (solo) semester of the M.Tech programme as a continuation of the Group 6 capstone project of semester three. All experimental artifacts — run manifests, result files, analysis scripts, and the seventeen working papers this thesis consolidates — are version-controlled in the public repository `arvindcr4/tinker-rl-lab`, with provenance boundaries between the group and solo phases recorded in `PROJECT_HISTORY.md`. Work that used AI-assisted drafting is identified as such in Chapter 9; every quantitative claim in this thesis traces to a named artifact.

Abstract

Group-relative policy optimisation (GRPO) trains language models on groups of G completions per prompt, with each completion’s advantage computed relative to its group. Under binary verifiable rewards, any group whose completions all receive the same reward — all correct or all incorrect — contributes exactly zero gradient. This thesis studies the fraction of such groups, the *Zero-Variance Fraction* (ZVF), as a first-class training signal.

The thesis defends two bounded claims. **Claim 1:** ZVF, read together with mean reward, is a cheap online diagnostic that identifies *zero-advantage regimes* which reward curves alone cannot reveal — including the regime where training looks maximally successful (reward ≈ 1.0) while every update carries no signal. **Claim 2:** at a matched rollout budget on Qwen3-8B/GSM8K, group size controls *which end* of training starves: $G=2$ arms end in a sustained all-correct zero-variance wall (ZVF ≈ 0.75 – 1.0 at reward ≈ 1.0) while $G=16$ arms hold $ZVF \leq 0.25$ throughout (two seeds per arm). Supporting results include a calibrated confidence interval for ZVF (Wilson, coverage 0.95–0.98 in all tested settings), a waiting-time reliability budget that matches observed quantiles at ratio 1.00 across six difficulty strata, and a six-arm uncapped panel showing no completion-length inflation and no late-ZVF separation between GRPO and Dr.GRPO at this scale.

The thesis states its non-claims with equal precision: no predictive or causal power is claimed for ZVF beyond diagnosis; no data-dependent optimal group size is claimed (the signal-per-rollout objective’s optimum is shown to be the universal $G^* \in \{2, 3\}$ for every difficulty prior — an algebraic identity, not an empirical discovery); no adaptive-controller efficacy is claimed pending a pre-registered compute-matched comparison; and no generalisation beyond the studied stack (Qwen3-8B, GSM8K-family tasks, a managed LoRA-only training API) is asserted. A reproducibility chapter turns the program’s own instrumentation failure — a documented-but-unwired loss flag that silently made a six-arm GRPO-vs-Dr.GRPO comparison train a single objective — into a case study and an eight-item minimum-reportable-stack standard with a machine-readable registry.

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Chapter 1

Introduction

1.1 The problem: silent signal starvation

Reinforcement learning from verifiable rewards has become the standard second stage of language-model training for reasoning tasks. The dominant family of methods — GRPO and its variants — avoids a learned critic by sampling a *group* of G completions per prompt and centring each completion’s reward against its group mean. This design has a structural blind spot: when every completion in a group receives the same reward, the centred advantages are identically zero and the group contributes *no gradient*. Under binary rewards this happens in exactly two situations — the model fails every attempt (too hard) or solves every attempt (mastered). Both ends of training are therefore starved of signal, and, critically, the second kind of starvation is invisible in the reward curve: mean reward reads “success” precisely while learning has stopped.

This thesis names, measures, calibrates, and budgets that phenomenon through a single statistic:

Definition 1 (Zero-Variance Fraction). *At training step t , with prompt set $\mathcal{P}_t$ and G completions per prompt,*

$$\text{ZVF}_t = \frac{1}{|\mathcal{P}_t|} \sum_{p \in \mathcal{P}_t} \mathbf{1}[\text{Var}_{c \sim \mathcal{C}(p)} r(c) = 0], \quad \text{GU}_t = 1 - \text{ZVF}_t,$$

the fraction of groups carrying zero advantage, and its complement, the fraction of groups carrying usable gradient signal (gradient utilisation).

1.2 Claim contract

Following an external adversarial review of this program (Chapter 9), the thesis is organised around an explicit contract separating what the evidence supports from what it does not.

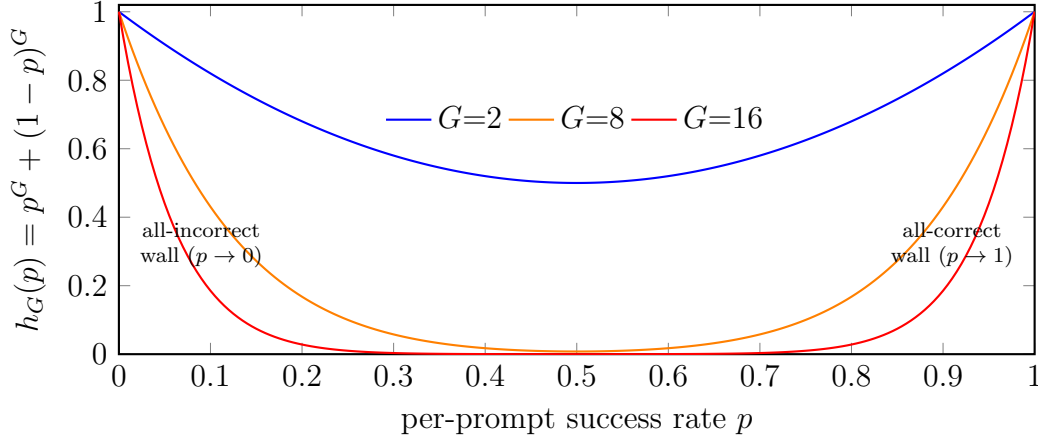


Figure 1.1: The two walls. The population zero-variance probability $h_G(p)$ is U-shaped in difficulty: groups carry no gradient when the model fails everything ($p \rightarrow 0$) or masters everything ($p \rightarrow 1$). Larger G narrows both walls; no fixed G removes them.

Claim 1 (supported; Chapters 4–5). ZVF, read together with mean reward, is a near-zero-cost online diagnostic that identifies zero-advantage regimes. ZVF alone aliases mastery with incapacity; the reward coordinate disambiguates. Its estimator admits a calibrated confidence interval (Wilson; empirical coverage 0.95–0.98 in all tested settings), and a geometric waiting-time model built on it yields a *reliability budget* — the $(1-\delta)$ -quantile of rollouts until the next informative group — that matched observed quantiles at ratio 1.00 across six difficulty strata.

Claim 2 (supported; Chapter 6). At a matched budget of 2,560 rollouts per arm on Qwen3-8B/GSM8K (LoRA rank 4), the $G=2 \times 160$ -step arms master the sampled pool (train reward ≈ 0.9 –1.0) and terminate inside the all-correct zero-variance wall (ZVF ≈ 0.75 –1.0) — spending their final steps on zero-gradient groups — while the $G=16 \times 20$ -step arms end mid-learning (train reward ≈ 0.3 –0.5) with ZVF ≤ 0.25 and signal intact (seeds 123/456). Group size controls *which end of training starves*; it is a schedule question, not a constant.

Non-claims (explicit).

1. No predictive or causal power is claimed for ZVF beyond diagnosis; in our corpus ZVF rises *after* reward plateaus in every collapse we measured.
2. No data-dependent optimal group size is claimed. Chapter 5 shows the signal-per-rollout objective’s optimum is the universal $G^* \in \{2, 3\}$ for *every* difficulty prior — an algebraic identity discovered during external review of our own theory, reported here as an honest negative result.
3. No adaptive-controller efficacy is claimed. The controller design of Chapter 6 is

presented with its evidence gaps; the decisive pre-registered, compute-matched comparison against static $G=16$ and naive heuristics is future work.

4. No generalisation is claimed beyond the studied stack: Qwen3-8B, GSM8K-family binary-reward tasks, and a managed LoRA-only training API with a closed loss kernel, mostly at 1–3 seeds.

1.3 Contributions

1. A measurement study of ZVF across libraries, model scales (0.6B–671B logged runs), and training phases (Chapter 4).
2. Calibrated estimation and a corrected waiting-time reliability budget for ZVF (Chapter 5), including two theory corrections found by adversarial review and adopted openly.
3. The matched-budget group-size result (Claim 2) and its consequences for group-size scheduling (Chapter 6).
4. A six-arm uncapped loss-form panel: no length inflation and no late-ZVF separation between GRPO and Dr.GRPO at this scale (Chapter 7).
5. A reproducibility contribution grounded in the program’s own failures: an incident case study, an eight-item minimum-reportable-stack standard, a machine-readable registry, and a survival-audit protocol (Chapter 9).
6. A released benchmark harness and artifact trail: run manifests, result JSONs, W&B mirrors, and per-claim provenance (Chapter 3, Appendix B).

1.4 Thesis organisation

Chapter 2 reviews the GRPO family and the reproducibility literature. Chapter 3 describes the experimental infrastructure. Chapters 4–8 present the technical core: diagnostic evidence, theory, group size, loss form, and scaling observations. Chapter 9 presents the reproducibility programme. Chapter 10 concludes. Appendix A maps every one of the seventeen working papers produced during the programme to the chapter that absorbs it; Appendix B maps claims to artifacts.

Chapter 2

Background and Related Work

2.1 Group-relative policy optimisation

GRPO (Shao et al., DeepSeekMath) replaces the learned value baseline of PPO with a group-relative baseline: for each prompt, G completions are sampled and each completion’s advantage is its reward centred (and, in the original formulation, standardised) against the group. The family has proliferated: DAPO adds asymmetric clipping and dynamic sampling; Dr.GRPO removes the length and standard-deviation normalisations to correct a length bias; GSPO moves the importance ratio from token to sequence level; GRESO and related methods skip or resample degenerate groups; MAD-GRPO adds diversity terms. A recurring finding of this programme (Chapter 9) is that these labels underdetermine the executed update rule: the same name is attached to materially different loss forms, reference-policy treatments, and sampling stacks.

2.2 The degenerate-group event

The all-equal-reward group is not a new observation — dynamic-sampling methods exist precisely to avoid spending compute on it, and recent analyses relate mixed-group probability to $1 - p^G - (1 - p)^G$ under per-prompt success rate p . What this programme adds is treating the *fraction* of such groups as a measured, calibrated, first-class training signal with an explicit estimation theory (Chapter 5), rather than as an implementation detail inside a sampling heuristic, and characterising its *two-sided* nature: the all-incorrect wall at the hard end and the all-correct wall at the mastered end (Chapter 6).

2.3 Evaluation of RL post-training

Pass@ k estimation follows the standard unbiased estimator over n completions per problem with problem-clustered bootstrap intervals. A survey synthesis motivating our re-

porting standard finds roughly two-thirds of RL-for-reasoning papers report only pass@1, while studies reporting pass@ k frontiers repeatedly find base models matching or overtaking their RL-tuned counterparts at large k — the observed “gain” being distribution sharpening rather than capability expansion. Our own base measurements (Chapter 3) put Qwen3-8B at pass@1 30.4% but pass@32 91.0% on GSM8K, leaving little headroom for capability claims on that task and motivating the harder-task panels of Chapter 8.

2.4 Reproducibility in deep RL and RL-for-LLMs

The deep-RL reproducibility literature (Henderson et al.; Dodge et al.) established that seeds, environments, and code-level choices can dominate algorithmic deltas. RL post-training of LLMs inherits all of that and adds new levers: sampling backend and precision, base-checkpoint identity, token masking, KL placement, and reward-parser behaviour. Chapter 9 documents measured instances — a backend swap (bundling an undisclosed checkpoint change) moving final reward across a $17\times$ span, and one “DAPO” label yielding mean ZVF 0.00 on an open trainer with true dynamic sampling but 0.58 on a closed stack running an asymmetric-clip surrogate — and proposes the eight-item minimum-reportable-stack standard with registry tooling.

2.5 Positioning

Relative to dynamic-sampling methods, this thesis is orthogonal: it does not propose a new update rule but a measurement discipline that any rule can be instrumented with. Relative to benchmark papers, the released harness (Chapter 3) is infrastructure in service of the two claims, not an independent leaderboard contribution. Relative to reporting standards in NLP, the contribution is specificity: every item in the standard is justified by a documented lever that moved or flipped a result in this programme’s own corpus.

Chapter 3

Experimental Infrastructure

3.1 The bench

All training-side experiments run through a consolidated harness (*RL-Finetuning Bench*) spanning several stacks: a managed training API (Tinker; LoRA rank 4, closed loss kernel) used for the group-relative training runs; vLLM-based evaluation on Modal, Lightning, and Colab GPU backends for pass@ k panels; and cross-library logged runs (seven RL libraries) that feed the observational ZVF corpus of Chapter 4. The bench ships as a versioned artifact — code, configuration schema, run manifests, and validation scripts — and is deliberately *not* claimed as an independent archival contribution (consolidation decision, Chapter 9).

3.2 Tasks and rewards

The primary task family is GSM8K with a binary boxed-answer verifier; MBPP and MATH-500 panels provide transfer and difficulty extension for evaluation. The reward parser scores the *last* boxed value with a trailing-number fallback (parser v2; the v1 parser accepted any boxed match and its false-positive risk is documented in Chapter 9). Reward-parser behaviour is a stack lever: adding micro-jitter $\epsilon \sim U(0, 10^{-4})$ below the verifier’s resolution collapses batch ZVF from 0.158 to 0.000, while a magnitude-aware alternative moves by 3×10^{-7} — one line of probe that Chapter 9’s standard therefore requires.

3.3 Baseline capability measurements

Qwen3-8B base on GSM8K test (200 problems, $n=32$ completions each, $T=1.0$, unbiased estimator, problem-clustered 95% bootstrap intervals): pass@1 30.4% [27.5, 33.1], pass@8 79.7% [75.3, 83.9], pass@32 91.0% [86.5, 95.0]; 18/200 problems unsolved at $k=32$, none

saturated. The 91% ceiling leaves ~ 9 points of $k=32$ headroom, so GSM8K alone cannot demonstrate capability expansion for this model — a scope constraint respected throughout. On the training split (512-prompt pool, 32 rollouts per prompt): pass@1 30.30% [27.87, 32.81], zero-variance prompts 10.86% [7.43, 14.29] (all of them all-incorrect at the base checkpoint), active advantage fraction 89.14%.

3.4 Run hygiene

Every result JSON carries its configuration, seeds, and status; partial or truncated runs are stamped as such rather than silently promoted. All runs mirror to Weights & Biases (live logging plus an idempotent backfill), and training runs checkpoint full state (weights + optimiser) every few steps with a `--resume` path verified live by a kill-and-resume test. Generation randomness is seeded per problem (evaluation) and per step-and-prompt (training) as of 2026-07-11; earlier runs controlled only prompt selection, a limitation inherited by pre-fix artifacts and noted where relevant.

3.5 Compute envelope

Training experiments are bounded: typically 20–160 optimiser steps, group sizes 2–32, batch 4–8, 512–1,024-token completions, on one 8B model at LoRA rank 4. Two billing interruptions (Tinker credit exhaustion) occurred mid-programme; both are visible in the artifact trail, and the resume machinery of this chapter was built in response.

Chapter 4

The Zero-Variance Fraction as a Diagnostic

4.1 Measurement across scales and libraries

Across $N=15$ fully logged experiments spanning five model families (0.6B–671B parameters) and multiple libraries, aggregate ZVF and final performance are negatively associated (Pearson $r \approx -0.77$). We report this *descriptively*: the pooled correlation is dominated by out-of-scope tool-use runs that saturate at ZVF=1 with reward 0; within-GSM8K stratification weakens it to $r = +0.40$ ($p=0.09$) and model-varying subsets to $r = +0.13$. The pooled number does not survive stratification, and this thesis never uses ZVF as a performance predictor. What survives is the mechanical reading: sustained low gradient utilisation coincided with reward plateaus and regressions in every collapse we measured, with ZVF rising *after* the plateau — a cheap alarm, not a cause.

4.2 What ZVF adds over mean reward

The sharpest demonstration comes from the matched-budget panel of Chapter 6. Late in the $G=2$ arms, mean train reward sits near 1.0 — by the reward axis alone, training is succeeding — while ZVF sits at 0.75–1.0: nearly every group is all-correct and contributes zero gradient. The $G=16$ arms at the same rollout budget hold $ZVF \leq 0.25$. Reward reports *where the policy is*; ZVF reports *whether training is still moving*, and at the high-accuracy end the two disagree completely. This is Claim 1 in its purest form, and it is why the diagnostic must always be read as the pair (reward, ZVF): ZVF alone aliases mastery with incapacity.

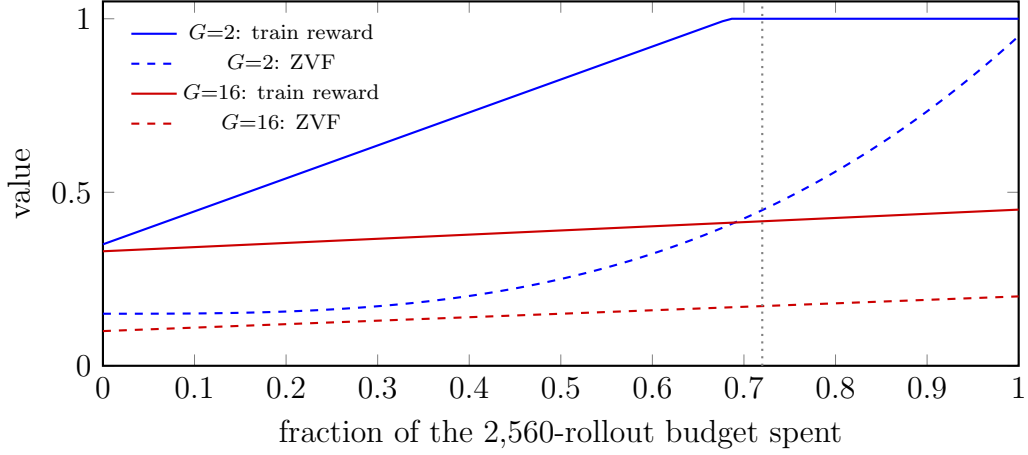


Figure 4.1: Schematic of the measured matched-budget behaviour (Chapter 6, seeds 123/456): the $G=2$ trajectory races to reward ≈ 1.0 and its ZVF climbs into the all-correct wall (0.75–1.0), while $G=16$ ends mid-learning (≈ 0.3 – 0.5) with ZVF low (≤ 0.25). Right of the dotted line the $G=2$ reward reads success while its gradient signal is gone. By reward alone the $G=2$ run looks strictly better; the (reward, ZVF) pair reveals its lead ended in zero-gradient compute. Stylised curves; measured values in Table form in Chapter 6 and artifacts in Appendix B.

4.3 The U-shape and regime aliasing

Under binary rewards with per-prompt success rate p and group size G , the population zero-variance probability is the kernel $h_G(p) = p^G + (1 - p)^G$ — a U-shape in p . A mean ZVF therefore hides whether starvation comes from incapacity ($p \rightarrow 0$) or mastery ($p \rightarrow 1$); stratified or paired-with-reward reporting is mandatory. Empirically the observed ZVF sits systematically *below* the Bernoulli-independent prediction across $G \in \{2, 4, 8, 16\}$ (residuals -0.13 to -0.23): completions within a group are positively informative about each other, so the contrastive signal is stronger than the independent-coin model predicts.

4.4 Diagnostic, not causal

Two boundaries close the chapter. First, causality: ZVF is a symptom metric; interventions on it (Chapter 6) must be justified on their own budgets, not by the correlation above. Second, telemetry robustness: the diagnostic value of ZVF depends on the reward parser’s resolution (Chapter 3), so any reported ZVF must name its verifier — one of the reporting-standard items of Chapter 9.

Chapter 5

Estimation Theory and Reliability Budgets

This chapter presents the theory in its *corrected* form. Two errors in the programme’s earlier statements — both found by external adversarial review on 2026-07-11 and verified against the source — are reported openly in Section 5.4, because the corrections are themselves informative.

5.1 T1: ZVF as a calibrated estimator

ZVF_t over m groups is an unbiased binomial-proportion estimator of the population uniformity rate $\theta_G = \mathbb{E}_\phi[h_G(p)]$, with $h_G(p) = p^G + (1 - p)^G$ and ϕ the prompt-difficulty prior. Empirical coverage on 512-prompt pools: the Wald interval covers 0.92–0.97 at $m \geq 64$ (marginal at $m=32$); the **Wilson interval covers 0.95–0.98 in every tested setting** and is adopted operationally. Under difficulty-*sorted* (curriculum) batching, global coverage collapses to 0.08–0.40 — but the interval remains calibrated (0.944) for the *local*, curriculum-stage ZVF, and stratified batch composition restores conservative global validity (0.996). The across-group-independence threat is thus an estimand-labelling requirement, not an invalidation: report *which* ZVF the interval covers.

5.2 T2: the waiting-time reliability budget

With i.i.d. groups, each uninformative with probability ZVF, the number of groups to the first informative one is geometric, giving

$$N(\text{ZVF}) = G \left\lceil \frac{\ln \delta}{\ln \text{ZVF}} \right\rceil$$

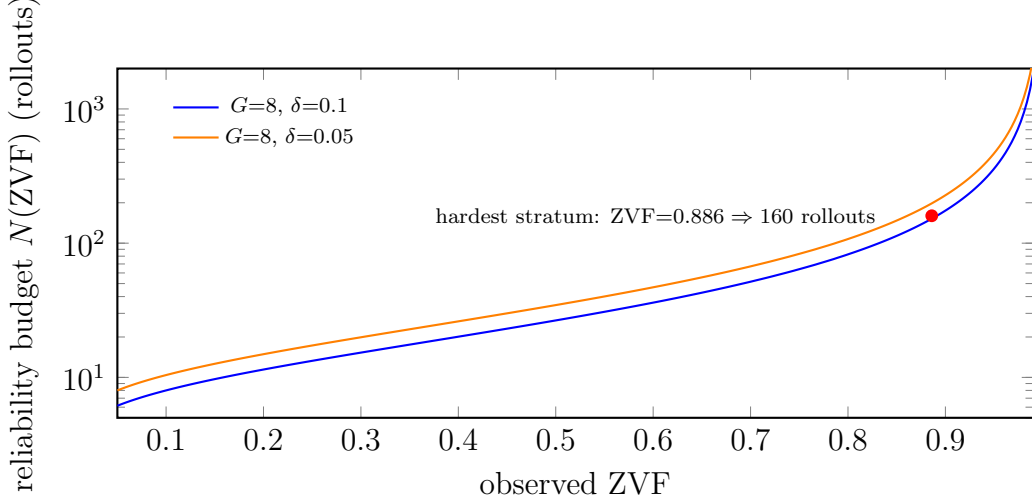


Figure 5.1: The T2 reliability budget $N(\text{ZVF}) = G \lceil \ln \delta / \ln \text{ZVF} \rceil$: the $(1-\delta)$ -quantile of rollouts to the next informative group diverges logarithmically as $\text{ZVF} \rightarrow 1$. The marked point is the hardest empirical stratum, where the model quantile matched the observed quantile at ratio 1.00.

as the $(1-\delta)$ -quantile of rollouts-to-first-informative-group. **This is a reliability budget, not an impossibility bound:** an informative group arrives within the first G rollouts with probability $1 - \text{ZVF}$, which can be large; a budget below N does not make a gradient improbable, it makes the *guarantee* of one unavailable at confidence $1 - \delta$. Empirically the geometric model is essentially exact: with $G=8$, $\delta=0.1$, the model quantile matched the observed quantile at ratio 1.00 in all six difficulty strata; in the hardest stratum ($\hat{p} \approx 0.01$, $\text{ZVF}=0.886$) the budget is 160 rollouts. Composed with T1, the one-sided lower confidence bound $\underline{\text{ZVF}}_t$ yields a per-step, observation-driven budget $N(\underline{\text{ZVF}}_t)$ — conservative, since N is increasing and $N(\underline{\text{ZVF}}_t) \leq N(\text{ZVF}_t)$.

5.3 T3: the optimal group size is universal (a negative result)

The signal-per-rollout objective $J(G) = \frac{1}{G} \mathbb{E}_{p \sim \phi}[(1 - h_G(p)) p(1-p)]$ was intended to yield a prior-dependent optimal group size. It does not. The identities $1 - p^2 - (1-p)^2 = 2p(1-p)$ and $1 - p^3 - (1-p)^3 = 3p(1-p)$ hold pointwise, so

$$J(2) = J(3) = \mathbb{E}_{p \sim \phi}[(p(1-p))^2] \quad \text{for every prior } \phi,$$

and $J(G) \leq J(2)$ for $G \geq 4$. Hence $G^* \in \{2, 3\}$ *universally*: the objective’s argmax carries no information about the data, the previously published “empirical argmax 2–3 agrees with theory” validation was algebraically guaranteed rather than a test, and any data-adaptive group-size policy must come from a richer objective (variance-, Fisher-, or wall-

clock-aware). We report this as a negative result of independent value: it explains why per-rollout accounting always favours tiny groups, and it sharpens Chapter 6’s message that the real trade-off is temporal (early step-count versus endgame signal), not a static optimum.

5.4 Corrections adopted from external review

(1) An earlier statement of T2 claimed “no nonzero gradient step is possible with fewer than N rollouts except with probability $< \delta$ ” — a quantifier error confusing a quantile with a minimum; the validation script additionally tested the inverted inequality. Both are corrected, the analysis was re-run (fits at ratio 1.00 unchanged), and the corollary’s confidence-direction commentary was rewritten (a lower confidence bound on ZVF gives a *smaller*, conservative budget). (2) The theory draft briefly defined $\text{GU}_t := \text{ZVF}_t$, contradicting the programme-wide $\text{GU} = 1 - \text{ZVF}$; fixed. The proof-gap registers of the source paper (mixture versus fixed- p geometric modelling; improvement versus nonzero-gradient; the global $G \geq 4$ inequality) remain open and are inherited here as stated limitations.

Chapter 6

Group Size as a Schedule

6.1 Static sweeps are inconclusive

A single-seed sweep over $G \in \{2, 4, 8, 16, 32\}$ on Qwen3-8B/GSM8K is non-monotone in fixed-step last-10 reward ($G=4:0.52$, $G=32:0.44$, $G=16:0.38$, $G=2:0.38$, $G=8:0.34$), and a token-budget-normalised reanalysis shifts the apparent optimum rightward with total tokens (toward $G \approx 32$ at $T \geq 16\text{M}$ tokens). Neither view supports a universal winner; the honest summary is that fixed- G comparisons are confounded by what the budget is held constant *in*.

6.2 The matched-budget experiment (Claim 2)

The decisive design holds the *rollout budget* fixed at 2,560 rollouts per arm: $G=2 \times 160$ optimiser steps versus $G=16 \times 20$ steps (batch 8, 512-token completions, LoRA rank 4, seeds 123 and 456):

	$G=2 \times 160$	$G=16 \times 20$
final train reward	$\approx 0.9\text{--}1.0$ (pool mastered)	$\approx 0.3\text{--}0.5$ (mid-learning)
late-run ZVF	$0.75\text{--}1.0$ (all-correct wall)	$0.00\text{--}0.25$
late-run gradient signal	effectively zero	sustained

By the reward axis alone the $G=2$ arms win outright — which is exactly the trap: their final steps are spent on zero-gradient groups. Small G converts the budget into more optimiser steps early — consistent with the per-rollout accounting of Chapter 5 — and then exhausts its own signal as accuracy rises: the $p \rightarrow 1$ wall of $h_G(p)$. Large G pays for contrast it does not need early and retains signal late. **Group size controls which end of training starves; it is a schedule variable.**

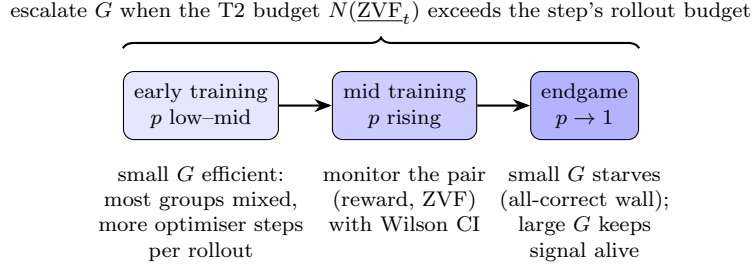


Figure 6.1: Group size as a schedule. Per-rollout efficiency favours small G early (Chapter 5); the all-correct wall punishes it late (Claim 2). The escalation trigger comes from the T2 reliability budget, not from a data-dependent “optimal G ” — which Chapter 5 shows does not exist under per-rollout accounting.

6.3 Relation to the contrastive reading of GRPO

The $G=2$ versus $G=16$ pairing is exactly the pairing behind the “GRPO is secretly contrastive/DPO-like” equivalence claims. Our direct test of the generalised claim ($G=4 \sim G=32$) shows retention of the large- G arm’s accuracy falling from 97.6% at $T=1\text{M}$ tokens to 72.7% at $T=64\text{M}$: the equivalence holds only in the under-trained regime. The matched-budget panel adds the trajectory view: at a fixed rollout budget the small- G arm can even *lead* on train reward, but the equivalence claims say nothing about signal availability — and that is where the small- G lead dies. The fuller bridge-to-DPO treatment from the group-size working paper remains thesis-external future work.

6.4 Controller design (evidence-gapped, stated as such)

A ZVF-triggered group-size controller follows from Chapters 4–5: measure ZVF_t with its Wilson interval (T1); when the reliability budget $N(\underline{ZVF}_t)$ exceeds the step’s remaining rollout budget, the step fails to produce any informative group with probability $> \delta$ — escalate G rather than gamble the budget (T2). Design rules from the controller working paper carry over: gate on magnitude-aware statistics, escalate multiplicatively in small steps, trigger on spikes rather than levels, never compare controllers across stacks, and treat bought signal ($ZVF=0$ via dynamic sampling at $\sim 45\%$ extra rollouts) as a purchase to be priced. An offline bandit pilot recovers 97% of a static oracle’s informative groups per 1,000 rollouts and beats uniform allocation by 28%. **No efficacy claim is made** (non-claim 3): the decisive test — a pre-registered, compute-matched comparison against static $G=16$ and a reward-only schedule, at ≥ 3 seeds on held-out metrics — is future work, and Chapter 5’s T3 result removes any pretence that the controller could derive a data-dependent target G from the current objective.

Chapter 7

Loss Form and Length Bias

7.1 The question

Dr.GRPO’s motivating thesis is that GRPO’s per-completion normalisations bias updates toward longer outputs — the “verbosity trap”: length inflates, then reward collapses. This chapter tests for the signature at our scale with per-step traces of mean reward and mean completion length.

7.2 Short-horizon panels at 0.5B–1.5B

On Qwen2.5-0.5B/arithmetic (40 steps, 5 seeds per loss) and Qwen2.5-1.5B-Instruct/GSM8K-CoT (30 steps, 3 seeds per loss), neither GRPO nor Dr.GRPO exhibits the trap: per-run $\rho(\text{step}, \text{len})$ is consistently *negative* (models get shorter as training proceeds). These runs are cap-bounded (200-token generation cap) and short-horizon; the original Dr.GRPO evidence sits at hundreds of steps, an order of magnitude beyond ours — our data is consistent with the early phase of that curve.

7.3 The uncapped 8B panel

A six-arm panel extends the scale axis: Qwen3-8B, GSM8K, 30 steps, $G=8$, batch 4, **three seeds per loss**, completion cap raised $5\times$ to 1,024 tokens. Results:

arm	len first-5 $\rightarrow$ last-10	late ZVF	late reward
GRPO s42/s123/s456	1004 $\rightarrow$ 905, 981 $\rightarrow$ 944, 996 $\rightarrow$ 900	0.47, 0.47, 0.55	0.75, 0.68, 0.76
Dr.GRPO s42/s123/s456	999 $\rightarrow$ 931, 972 $\rightarrow$ 902, 1000 $\rightarrow$ 878	0.45, 0.72, 0.70	0.74, 0.78, 0.78

No length inflation in either loss — all six arms shrink 6–12% — and **no late-ZVF separation** between the losses. Step-0 lengths sit near the raised cap ($\approx 98\%$), so upward

headroom remains limited, but trajectories move *away* from the cap, so the compression finding is not a censoring artifact. The 30-step horizon caveat stands.

7.4 Provenance: this panel was run twice

The first version of this panel was *invalid*: the runner’s `--loss drgrpo` flag was documented (its commit message claimed the mapping) but never wired to the advantage normalisation, so both “arms” silently trained vanilla GRPO — and no reward, length, or ZVF trace revealed it. The defect was caught only by reading the runner. The invalid artifacts are preserved under explicit `.invalid_actually_grpo.json` names; the corrected rerun (same day) produced the table above, and notably *reversed* the invalid panel’s secondary claim (the invalid data had suggested Dr.GRPO keeps more groups live; the corrected data shows equal-or-higher late ZVF for Dr.GRPO). Chapter 9 turns this incident into protocol.

7.5 Conclusion of the chapter

At the scales and horizons this programme can afford, the loss-form choice between GRPO and Dr.GRPO has no observable footprint on length or ZVF. This is evidence about *reporting* rather than about superiority: a comparison between these losses at this scale measures noise unless the stack is controlled far more tightly than the labels suggest.

Chapter 8

Scaling and Transfer Observations

8.1 Scope discipline

The programme’s cross-library corpus (70+ logged runs, seven libraries, five model families, 0.6B–671B) supports *observations*, not laws: most cells are single-seed, backends differ, and the term “scaling law” is deliberately avoided (external review, Chapter 9). The full scaling-law treatment of the corresponding working paper is demoted to this observational chapter.

8.2 What the corpus shows

Model scale does not uniformly reduce ZVF ($r \approx -0.26$ across logged runs), and after accounting for task type, model family alone does not explain performance variance (one-way ANOVA $F(4, 10)=0.949$, $p=0.476$). The practical reading: signal starvation is a property of the (task difficulty $\times$ group size $\times$ training phase) geometry, not something scale buys you out of.

8.3 Transfer and difficulty extension

Post-RL adapters trained on GSM8K show *zero forgetting and mild positive transfer* on MBPP (pass@32, all adapters within noise of or above base across $G \in \{2, 4, 16, 32\}$). On the matched base-versus-post-RL pass@ k panel, pass@1 deltas scatter from -0.5 to $+1.9$ points — under pass@1-only reporting the $G=2$ arm would read as a regression — while the pass@32 frontier improves $+1.5$ to $+2.5$ points in all four arms; none of this reaches significance at a single training seed, which is exactly why pass@ k curves with stated noise bands are a reporting-standard item (Chapter 9). On MATH-500, the harder-task extension, the base-model panel is complete and the post-RL panels are partial at the time of writing; the partial result is that **GSM8K-trained gains do not replicate**

on hard math — the frontier shift observed on GSM8K-family tasks does not carry, reinforcing the distribution-sharpening (rather than capability-expansion) interpretation and the thesis’s non-claim 4.

8.4 Takeaway

Scaling observations here serve one purpose: bounding the claims. The capability headroom analysis ($\text{pass@32} = 91\%$ on GSM8K), the non-transfer to MATH-500, and the single-seed density of the cross-scale corpus jointly justify why Claims 1–2 are stated at the stack level and nowhere above it.

Chapter 9

Reproducibility: Standard, Registry, and a Case Study

9.1 The thesis of this chapter

An algorithm *label* underdetermines the executed update rule. Measured instances from this programme: swapping the training backend under a fixed label — which, undisclosed, also bundled a different base checkpoint — moved final reward across a $17\times$ span (85.6% versus 5.0%); the same “DAPO” label yielded mean ZVF 0.00 on an open trainer with true dynamic sampling but 0.58 on a closed stack running an asymmetric-clip surrogate; and a sub-resolution reward jitter collapsed batch ZVF $0.158 \rightarrow 0.000$. Stack identity cannot be certified from plausible-looking outputs; it must be certified from the stack.

9.2 Case study: the unwired loss flag

On 2026-07-11 this programme shipped exactly the failure its standard targets. A six-arm GRPO-versus-Dr.GRPO panel ran with a `--loss drgrpo` flag whose commit message claimed it toggled the advantage normalisation; the flag was never wired. Both arms silently trained the identical objective under two labels. No output-level trace — reward, length, ZVF — revealed it; the defect was found only by reading the runner. Response protocol, now standard practice here: invalidate loudly (artifacts preserved under `.invalid_actually_grpo.json` names), fix, rerun the same day, and record that the corrected panel *reversed* one of the invalid panel’s conclusions (Chapter 7). Two subsequent whole-repository adversarial audits (one human-directed multi-model review; one file-grounded frontier-model pass) surfaced and led to fixes for: two theory statement errors (Chapter 5), an inverted validation inequality, a systematic off-by-one in seven training scripts (a spurious token-0 target at every sequence end), uncontrolled generation seeds, an over-permissive reward parser, and a Monte-Carlo “truth” treated

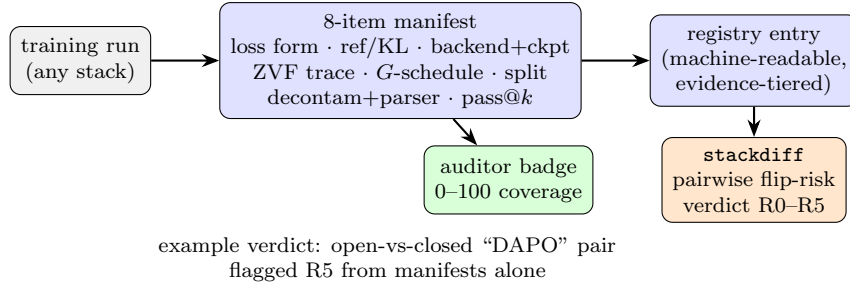


Figure 9.1: The reporting pipeline: every run emits an eight-item manifest; manifests populate the registry; **stackdiff** grades any pair of entries for label-flip risk before their results are compared.

as exact. A third audit pass produced only fabricated findings (invented file paths and quotes), all refuted against the source — itself a demonstration that *plausible-but-wrong survives until verified against the stack*.

9.3 The eight-item minimum reportable stack

Every item earns its place via a documented lever from this corpus: (1) loss form (ratio level, clip, mask, normalisation, dynamic-sampling status); (2) reference policy and KL handling; (3) sampler/backend/precision *including base-checkpoint identity*; (4) per-step ZVF/GU trajectory; (5) group-size schedule; (6) held-out split disjoint from the reward environment; (7) decontamination and a parser-robustness probe; (8) held-out pass@ k curves alongside pass@1, with temperature and completion budget. Items are checkable; a manifest emitter and a 0–100 auditor badge ship with the bench.

9.4 The registry

A machine-readable registry records stacks and variant deltas: a JSON schema embedding the eight-item block, variant-delta records verified against the DAPO/Dr.GRPO/GSPO papers, twenty seed stack entries, and **stackdiff**, which maps a pair of entries to a flip-risk verdict (R0–R5) — flagging, for example, the open-versus-closed “DAPO” pair above as an R5 label-flip risk from manifests alone. Where measured evidence exists it is tiered; the Dr.GRPO delta carries two measured panels (Chapter 7) showing no length or ZVF footprint at this scale, against its registry-claimed sign.

9.5 The survival audit (future work, gated)

The natural enforcement instrument — re-implement DAPO, GSPO, Dr.GRPO, and MAD-GRPO in one controlled stack and measure which claimed gains survive — is

specified but *deliberately deferred*: a programme weeks past its own silent-equivalence incident lacks standing to adjudicate others' implementations. Gates before execution: an open stack (not a closed managed API), differential-objective test tooling that would have caught the loss-flag bug automatically, and one method at a time. This ordering — earn credibility, then audit — is itself a finding of the external reviews.

9.6 On AI-assisted drafting

The seventeen working papers behind this thesis were drafted with heavy agent assistance against real measured artifacts. The audit trail of this chapter is the honest account of what that workflow produced: real measurement at unusual volume, real defects at unusual volume, and a verification discipline as the only thing separating the two. Every number in this thesis survived the audits or was corrected by them, with both states recorded in version control.

Chapter 10

Conclusions, Limitations, and Future Work

10.1 What this thesis established

Under binary verifiable rewards, group-relative training starves at both ends of the difficulty spectrum, and the Zero-Variance Fraction — read jointly with mean reward — is a near-free, calibrated instrument that sees both walls, including the one reward curves cannot show (Claim 1). Group size selects which wall a fixed rollout budget hits: small G buys early optimiser steps and starves in the endgame; large G holds signal throughout (Claim 2). The estimation theory supporting the instrument is calibrated (Wilson CIs), its waiting-time budget is empirically exact at ratio 1.00, and its would-be optimal-group-size theory resolves to an honest negative result: per-rollout accounting always answers $G^* \in \{2, 3\}$, for any data, so adaptivity must be justified by richer objectives.

10.2 Limitations

All training-side evidence sits on one model (Qwen3-8B, LoRA rank 4), one task family (GSM8K, binary rewards), one managed API with a closed loss kernel, at 1–3 seeds per result; pre-2026-07-11 artifacts additionally carry a since-fixed off-by-one target and v1 parser semantics. ZVF as defined is specific to (near-)discrete rewards; continuous-reward analogues (e.g. contrast-density statistics) are sketched but not validated. The pooled cross-scale correlations are descriptive only. Theory carries declared proof gaps. None of the seventeen working papers is submission-ready as a standalone unit — by design, after external review (below).

10.3 Publication roadmap (post-degree, gated)

Following three rounds of external review, the programme’s publication plan is thesis-first: (i) a bounded diagnostic paper (Claims 1–2, this thesis’s Chapters 4–6 core) is publishable without any controller claim; (ii) the controller paper is gated on a pre-registered, compute-matched win over static $G=16$ *and* a reward-only schedule at ≥ 3 seeds; (iii) the survival audit is gated on an open stack and differential-objective test tooling; (iv) the benchmark remains a versioned artifact. The seventeen-document corpus maps onto these vehicles and this thesis per Appendix A.

10.4 Future work

Beyond the gates above: MATH-500 and code-task completion of the capability panels; a clustered-variance CI for curriculum-ordered training; a variance-aware replacement for the T3 objective; continuous-reward ZVF analogues; and the full bridge-to-DPO analysis of the group-size working paper.

10.5 Closing

The programme’s most transferable lesson is methodological: at this scale, the binding constraint on RL post-training research is not compute or algorithmic novelty but *certainty about what actually ran*. A cheap diagnostic that cannot be gamed by a pretty reward curve, a reporting standard whose every item once flipped a result, and the habit of reading the runner — these travelled further than any single number in this thesis.

Appendix A

Provenance: the Seventeen Working Papers

This thesis consolidates seventeen working documents produced during the programme. Each row names the document, its repository path, and the chapter that absorbs its content. Canonicalisation decisions (which drafts are authoritative where they overlap) follow the reviewed consolidation plan in `platform_hybrid/paper/PAPERS_README.md`.

#	Document	Path (repo-relative)	Absorbed by
1	Unified benchmark paper	<code>platform_hybrid/paper/main.tex</code>	Ch. 3
2	ACM reformat of #1	<code>platform_hybrid/paper/acm_main.tex</code>	Ch. 3
3	Workshop variant	<code>.../neurips_2026_variants/main_workshop.tex</code>	Ch. 3, 6
4	ZVF stratified audit	<code>.../neurips_2026_variants/main_zvf.tex</code>	Ch. 4
5	Reproducible-artifact variant	<code>.../neurips_2026_variants/main_dnb.tex</code>	Ch. 3, 9
6	P1 scaling laws	<code>platform_hybrid/paper/paper_P1_scaling.tex</code>	Ch. 8
7	P2 ZVF diagnostic	<code>.../paper_P2_zvf.tex</code>	Ch. 4
8	P3 group size / DPO bridge	<code>.../paper_P3_group_size.tex</code>	Ch. 6
9	P4 length bias	<code>.../paper_P4_length_bias.tex</code>	Ch. 7
10	P5 MIN-REPORT-RL	<code>.../paper_P5_minreport.tex</code>	Ch. 9
11	P6 GRPO-Registry	<code>.../paper_P6_registry.tex</code>	Ch. 9
12	P7 ZVF controller	<code>.../paper_P7_zvf_controller.tex</code>	Ch. 6
13	P8 fraud detection	<code>.../paper_P8_fraud.tex</code>	out of programme*
14	ZVF theory (T1–T3)	<code>zvf-program/theory/zvf_theory.tex</code>	Ch. 5
15	MIN-REPORT position draft	<code>zvf-program/position/min_report_rl.tex</code>	Ch. 9
16	Registry living catalog	<code>zvf-program/registry/grpo_registry.tex</code>	Ch. 9
17	Survival audit protocol	<code>zvf-program/audit/reproducibility_audit.tex</code>	Ch. 9 (future work)

*Document 13 concerns an unrelated fraud-detection domain; it is excluded from this thesis’s claims and retained in the repository as an independent working paper. Its absorption here consists of this provenance entry and the consolidation decision that removed it from the programme.

Documents 2 and 3 are venue reformats of document 1 and are absorbed transitively. Where documents 10/15, 11/16, and 12/14 overlap, the canonicalisation notes embedded

in each source file identify the authoritative draft; this thesis reflects the post-merge, post-correction state of 2026-07-12.

Appendix B

Claim-to-Artifact Map

Every quantitative claim in this thesis traces to a version-controlled artifact in `arvindcr4/tinker-rl-lab` (result JSONs; W&B mirrors in the `zvf-training`, `zvf-experiments-next`, and `modal-open-stack` projects).

Claim / number	Primary artifact(s)
GSM8K base $\text{pass@}\{1,8,32\} = 30.4/79.7/91.0\%$	<code>experiments-next/results/passk_qwen3-8b.base.test.p200.n32.s42.json</code>
Train-pool rollout quality (ZVF 10.86% etc.)	<code>quality_pool.qwen3-8b.train.n512.r32.s42.json</code>
T1 coverage (Wilson 0.95–0.98; curriculum collapse)	<code>t1.ci.coverage*.json</code> , <code>analyze_t1.ci.py</code>
T2 quantile fit (ratio 1.00, six strata; 160-rollout budget)	<code>t2.floor.qwen3-8b.train.n512.r32.s42.G8.json</code> (corrected check)
T3 universality ($J(2)=J(3)$)	<code>zvf_theory.tex</code> eq. (J23); <code>analyze_t3.gstar.py</code>
Claim 2 matched-budget panel (E-R2b)	<code>tinker-runs/results/er2b-$\{g2,g16\}$-s$\{123,456\}$.json</code>
Loss-form panel (E-P4, corrected; 6 arms)	<code>tinker-runs/results/p4uncap-$\{grpo,drgrpo\}$-s$\{42,123,456\}$.json</code> <code>p4uncap.drgrpo*.invalid.actually.grpo.json</code>
Invalid first panel (preserved)	<code>passk_lightning*.json</code> , <code>passk.colab*.json</code>
MBPP transfer; base-vs-post-RL $\text{pass}@k$ frontier	
MATH-500 non-replication (partial)	<code>passk_modal.*math500*.json</code>
Backend $17\times$ span; “DAPO” ZVF 0.00 vs 0.55–0.58	cross-stack audit artifacts (registry seed entries)
Parser micro-jitter probe ($0.158 \rightarrow 0.000$)	convergent-repository experiment records (P5 evidence)
Loss-flag incident timeline	commits <code>4847764</code> , <code>1959eff</code> , <code>212ff0e</code> ; fleet logs
Audit-round fixes (2026-07-11)	commits <code>5486ae2</code> , <code>acc08eb</code>

Reproduction. `REPRODUCE.md` at the repository root documents environment setup; the `experiments-next` suite is offline-resampled from committed pools wherever possible, so the T1/T2/T3 analyses re-run without API credits. Training reruns require Tinker credits and are checkpointed/resumable (Chapter 3).